University of St.Gallen

Course and Examination Fact Sheet: Autumn Semester 2025

7,005: RPV: Prompt Engineering: Innovation through generative AI

ECTS credits: 4

Overview examination/s

(binding regulations see below)
decentral - Software application, Digital, Group work group grade (50%)
Examination time: Term time
decentral - Written work, Digital, Group work group grade (50%)
Examination time: Term time

Attached courses

Timetable -- Language -- Lecturer
7,005,1.00 RPV: Prompt Engineering: Innovation through generative AI -- English -- Rietsche Roman

Course information

Course prerequisites

Important: This course is designed for both students with and without programming skills.

No prior knowledge of programming languages like Python is necessary for the course. However, if you are already familiar with programming, you can utilize the Python libraries of openAI or similar, using models such as chatGPT, GPT4, etc. (for example, Langchain). On the other hand, if you are unfamiliar with programming, you can use low-code engines that facilitate a drag-and-drop approach (for example, Langflow).

Your grade is independent of your programming skills and depends primarily on the ideas and work you have done on the project.

The course aims to go beyond the chatGPT user interface and explore using APIs (Application Programming Interfaces) to integrate large language models and generative AI into your own applications.

This course is assigned to the Transforming and Managing Digital Business profile, but can also be taken without selecting a specialisation.

No strict requirements, but helpful:

https://www.deeplearning.ai/

https://www.deeplearning.ai/short-courses/collaborative-writing-and-coding-with-openai-canvas/

https://www.deeplearning.ai/short-courses/reasoning-with-o1/

https://www.deeplearning.ai/short-courses/ai-python-for-beginners/

https://www.deeplearning.ai/courses/generative-ai-for-everyone/

https://www.coursera.org/learn/ai-for-everyone

Learning objectives

1. You will understand and be able to explain the fundamentals of generative AI and outline applications of generative AI, focusing on natural language processing and prompt engineering.
2. You will be able to create your own solutions using generative AI models and prompt engineering, demonstrating your problem-solving and design skills.
3. You will be able to identify and critically analyze generative AI's potential biases and ethical implications.
4. You will be able to work with industry partners to implement AI solutions, demonstrating teamwork and practical application.
5. You will be able to document and present project results according to scientific standards and effectively communicate technological innovations.

Course content

In a nutshell

You will ...

- Work on a **real-world challenge** presented by an industry partner (we partner with multiple **start-ups/larger organizations**), which will involve applying the concepts of generative AI and prompt engineering learned in the course to design an effective solution.
- Engage in an iterative process to design and develop a Minimum Viable Product (MVP), gaining practical experience and feedback from your coach that will help refine your solution for optimal performance and usability in the field.
- Apply prompt engineering principles to direct your problem-solving methodology, and implement an evaluation process to assess the effectiveness of your created artifact in real-world scenarios.
- Present your solution and the insights to the industry partner, demonstrating your communication skills and capacity to deliver AI solutions in a real-world context.

As Artificial Intelligence progresses, generative models, including chat models, emerge as transformative tools across various industries. From fostering creativity through content generation to fortifying decision-making processes, these models have ignited a new era of technological innovation. In this Master's course, we delve into the exciting domain of generative AI, focusing on the art of prompt engineering and its practical applications in chat models and beyond.

Structured as a Research and Practice Venture, the course immerses students in hands-on, industry-based projects. In collaboration with industry partners, participants will address real-world challenges by adeptly leveraging generative AI, particularly crafting and fine-tuning prompts for chat models. With a blend of theory, practice, and collaboration, this course serves as a launchpad for aspiring innovators seeking to sculpt the future with generative AI and chat models.

Course structure and indications of the learning and teaching design

This course counts 4 credits, corresponding to a total workload of 120 hours. This includes self-study, campus time, and all examinations. The course will be conducted in person in St.Gallen.

The course is structured as follows:

- **Kick-Off:** In the introductory session, students will receive an overview of Generative AI, its applications, and its potential in innovation. The session will also cover the introduction to the industry projects and the course structure.
- **Methods Course:** This session lays the foundation for working on the projects by providing students with the methodologies and tools needed for prompt engineering and development with Generative AI.
- **Interim Presentation (not grading relevant):** Halfway through the processing phase, students will present their progress and receive feedback from coaches and fellow students. This is an opportunity to refine their approach and implement feedback for the final project.
- **Final Presentation (grading relevant):** Students will present their completed projects to industry partners and peers. This session serves as an avenue to showcase the innovation achieved through Generative AI and receive feedback.

In between these sessions, students will work on their projects in groups of 3-4. They will have regular meetings with coaches who will provide feedback and input to the student groups.

The final deadline for the submission of all documentation (grading relevant) corresponds with the end of the official lecture period.

Course literature

Literature on the various project topics will be presented at the introductory session (some helpful papers are):

- Training Language Models to Follow Instructions with Human Feedback (2022) by Ouyang, Wu, Jiang, Almeida, Wainwright, Mishkin, Zhang, Agarwal, Slama, Ray, Schulman, Hilton, Kelton, Miller, Simens, Askell, Welinder, Christiano, Leike, and Lowe https://arxiv.org/pdf/2203.02155.pdf
- Fine-Tuning Language Models from Human Preferences (2020) by Ziegler, Stiennon, Wu, Brown, Radford, Amodei, Christiano, Irving https://arxiv.org/abs/1909.08593
- Learning to Summarize from Human Feedback (2022) by Stiennon, Ouyang, Wu, Ziegler, Lowe, Voss, Radford, Amodei, Christiano https://arxiv.org/abs/2009.01325
- How Close is ChatGPT to Human Experts? Comparison Corpus, Evaluation, and Detection by Biyang Guo, Xin Zhang, Ziyuan Wang, Minqi Jiang, Jinran Nie Yuxuan Ding, Jianwei Yue, Yupeng Wu https://arxiv.org/pdf/2301.07597v1.pdf
- Language Models are Few-Shot Learners by Tom B. Brown et al https://arxiv.org/pdf/2005.14165.pdf

Additional course information

Mr. Andreas Göldi (andreas.goeldi@unisg.ch) and Prof. Dr. Roman Rietsche (roman.rietsche@unisg.ch) will be happy to answer any questions you may have.

Examination information

Examination sub part/s

1. Examination sub part (1/2)

Examination modalities

Examination type	Software application
Responsible for organisation	decentral
Examination form	Practical test
Examination mode	Digital
Time of examination	Term time
Examination execution	Asynchronous
Examination location	On Campus
Grading type	Group work group grade
Weighting	50%
Duration	--

Examination languages

Question language: English
Answer language: English

Remark

Presentation and evaluation of developed prototype

Examination-aid rule

Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids

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2. Examination sub part (2/2)

Examination modalities

Examination type	Written work
Responsible for organisation	decentral
Examination form	Written work
Examination mode	Digital
Time of examination	Term time
Examination execution	Asynchronous
Examination location	Off Campus
Grading type	Group work group grade
Weighting	50%
Duration	--

Examination languages

Question language: English
Answer language: English

Remark

Scientific Paper describing the prototype

Examination-aid rule

Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids

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Examination content

The content of the presentation depends on your project.

For the scientific paper the basic structure will be:

1. Introduction and Problem Statement:
    - You will be expected to provide a clear and concise introduction to the problem to be solved in your project.
    - Explain the significance of the problem and its relevance to generative AI.
    - Clearly define your contribution
2. Theoretical Background:
    - This section will test your understanding of the theoretical foundations of generative AI.
    - Discuss the fundamental concepts, techniques related to generative models and prompt engineering.
    - Explain the advantages and limitations of different approaches within generative AI.
3. Methodology:
    - Present a detailed account of the methodologies you employed in your project.
    - Describe the data collection process, preprocessing techniques, and how you used generative AI models.
    - Justify your choices and demonstrate a deep understanding of the underlying principles.

4. Results and Discussion:

   - This section will evaluate your ability to analyze and interpret the outcomes of your project.
   - Present the results of your experiments and analyses in a clear and organized manner.
   - Discuss the strengths, weaknesses, and implications of your findings.
   - Reflect critically on your work and propose potential areas for improvement.

Examination relevant literature

No special literature required.

Please note

Please note that only this fact sheet and the examination schedule published at the time of bidding are binding and takes precedence over other information, such as information on StudyNet (Canvas), on lecturers' websites and information in lectures etc.
Any references and links to third-party content within the fact sheet are only of a supplementary, informative nature and lie outside the area of responsibility of the University of St.Gallen.

Documents and materials are only relevant for central examinations if they are available by the end of the lecture period (CW51) at the latest. In the case of centrally organised mid-term examinations, the documents and materials up to CW 43 (Freitag, 24. October 2025) are relevant for testing.

Binding nature of the fact sheets:

- Course information as well as examination date (organised centrally/decentrally) and form of examination: from bidding start in CW 34 (Thursday, 21 August 2025);
- Examination information (supplementary aids, examination contents, examination literature) for decentralised examinations: in CW 42 (Monday, 13. October 2025);
- Examination information (supplementary aids, examination contents, examination literature) for centrally organised mid-term examinations: in CW 43 (Freitag, 24. October 2025);
- Examination information (regulations on aids, examination contents, examination literature) for centrally organised examinations: two weeks before ending with de-registration period in CW 45 (Monday, 03. November 2025).